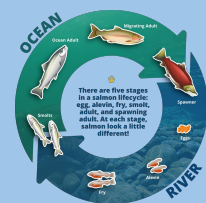




Salmon Populations and Climate Change

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Introduction:

For thousands of years, salmon have been the life force of California's rivers nourishing entire ecosystems, sustaining Tribal cultures, and anchoring local economies. Today, that legacy stands on the brink. A century of dam building, habitat loss, and accelerating climate change has pushed these iconic fish toward collapse, turning their struggle into a clear warning about the health of our waterways. Yet amid the crisis, bold restoration efforts from historic dam removals to renewed tribal stewardship offer a powerful glimpse of hope. Understanding this turning point is essential to shaping policies that can determine whether salmon disappear, or once again thrive in California's rivers.

Background:

Pre-European: Salmon and acorns made up more than ½ of the Karuk people (along the Klamath)

Late 1800s/Early 1900s: hydraulic mining ruined river habitats, dams blocked access to high elevation streams with cooler water & breeding grounds, commercial catch of Chinook dropped from 11 million lbs in 1880 to under 3 million lbs in 1939

Late 1900s: 1966-1980 commercial catches of Chinook were ~550,000, Sacramento river winter run adult salmon decreased from 100,000 (in the early 1980s) to less than 200

2008: Chinook fishery was closed due to the collapse of the Sacramento river run causing salmon to not survive their journey

General threats: Vulnerable to hotter temperatures, drying up of streams/river that are used for breeding. Human activity (water redirection/pollution) compounds on climate change to reduce salmon access to healthy breeding grounds leading to smaller populations.

Main Findings:

Dam Removal: Taking down the Klamath dams opened up around 420 miles of old habitat and let salmon back into colder upstream waters, which has helped reduce warm water build-up and algae problems. Due to it, better migration timing is already visible.

Hatcheries: They still help keep salmon numbers from crashing, but they can water down wild genetics and cause extra competition. The current idea is to use hatcheries carefully and track better, not rely on them forever.

Genetics & Adaptation: Breeding salmon to handle more heat is not as effective and hits a limit, thus the main strategy is just protecting the natural diversity they already have and keeping access to cold-water areas.

Climate Warming: Rising river temps in summer make migration harder and reduce spawning success. Salmon that can reach higher elevations with colder water do better, thus keeping those pathways open are crucial.

Policy implications:

Climate Resilience: Restoring habitats and removing dams builds resilience to hotter, drier conditions and prevents ecosystem collapse.

Tribal Healing & Culture: Salmon restoration honors Tribal Nations' millennia-old cultures and supports food security and heritage for all Californians.

Economic & Water Benefits: Protects fisheries, tourism, recreation jobs, and drives water reforms that keep streams cold and ecosystems healthy.

Urgency: Without immediate action, salmon face extinction and cultural/economic losses become irreversible

Opposing Arguments:

- ❖ GMO/breeding can quickly "climate-proof" salmon and sustain fisheries
 - Rebuttal: This harms natural genetics and fails when temperatures rise further.
- ❖ Dams = carbon-free power
 - Rebuttal: Targeted dams produce <2% of power and lose far more long-term water security than they save; removal is a gain.

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SPECIES	AVERAGE 2017 SCORE	AVERAGE LEVEL OF CONCERN
Chinook (8 kinds)	2.3	HIGH
Coho (2 kinds)	1.5	CRITICAL
Chum (1 species)	1.6	CRITICAL
Pink (1 species)	1.6	CRITICAL